finite-dim vector space

has spanning list of vectors (can be reduced to basis)

- list of vectors $v_1, \dots, v_m \in V$ (V is finite-dim)

(linear combination)

① spanning list $\{a_1 v_1\} + \dots + \{a_m v_m\} = V$ (smallest containing subspace)

② independent $\{a_1 v_1\} \oplus \dots \oplus \{a_m v_m\}$

③ basis $\{a_1 v_1\} \oplus \dots \oplus \{a_m v_m\} = V$

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dependent $v_1, \dots, v_m$ then $v_1 \dots v_{k-1}\ v_k \dots v_m$

linear combination of prev v's

$$\overbrace{\overbrace{v_1 \dots v_{k-1}}^{\text{independent}}\ w_k \dots \cancel{w} \dots w_m}^{\text{spanning}}$$

v_k

linear combination of prev vectors

finite-dim space length comparison

$$\text{independent} \le \underbrace{\text{basis}}_{= \dim V} \le \text{spanning}$$

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finite-dim space

spanning list $\xrightarrow{\text{reduce}}$ basis remove $\cancel{v_k} \in \operatorname{span}(\overbrace{v_1 \dots v_{k-1}}^{\text{independent}})$

independent $\xrightarrow{\text{extend}}$ basis reduce $\overbrace{u_1 \dots u_m}^{\text{independent}}\ \overbrace{w_1 \dots w_m}^{\text{spanning list}}$

Note. independent / spanning list

with length $= \dim V$ is basis (cannot reduce)

U is subspace of V then

① $\dim U \le \dim V$

② $U \oplus W = V$ (subspace W)

③ if $\dim U = \dim V$ then $U = V$

$\because$

$$\overbrace{\underbrace{u_1, \dots, u_m}_{\text{basis of } U} \; w_1, \dots, w_n}^{\text{basis of } V}$$

extended

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$$

if V_1, V_2 subspace of finite-dim vector space

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basis of $V_1 \cap V_2$: $v_1, \dots, v_n$

basis of V_1 : $v_1, \dots, v_n, u_1, \dots, u_j$ (extend)

basis of V_2 : $v_1, \dots, v_n, w_1, \dots, w_k$ (extend)

basis of $V_1 + V_2$: $v_1, \dots, v_n, u_1, \dots, u_j, w_1, \dots, w_k$

$\because$ otherwise non-trivial

$$\begin{aligned} 0 = \; & \left.\begin{aligned} & a_1 v_1 + \dots + a_n v_n \\ & + b_1 u_1 + \dots + b_j u_j \end{aligned}\right\} \in V_1 \\ & \left. + c_1 w_1 + \dots + c_k w_k \right\} \in V_2 \text{ and } V_1 \end{aligned}$$

however $d_1 v_1 + \dots + d_n v_n + c_1 w_1 + \dots + c_k w_k = 0$

only have trivial solution